

Analog IC Design Challenge: Self-Biased Sub-1V Bandgap Reference

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Technology	65nm
Supply voltage	1.2
Corners	Temp: -40 to 125 Process: SS, SF, FS, FF
Output voltage	0.8
Total bias current	< 10uA
Phase margin	> 60°

1 Part 1: BGR Core and Startup Circuit (Behavioral Amplifier)

1.1 Hand Analysis

Hand analysis

$$\frac{\alpha_1}{R_2} = \frac{b_1}{R_1} \quad \alpha_1 = \frac{k}{2} \ln$$
$$b_1 \approx 1.5 - 2 \text{ mV}$$

I choose $b_1 = 1.75 \text{ mV}$
calculate $\alpha_1 = \frac{k}{2} \ln$

$$\frac{R_1}{R_2} = \frac{b_1}{\alpha_1} = 6.39$$
$$R_1 = 6.39 R_2$$

I choose $1 \mu\text{A}$ to pass in the branch of mosfet
So

$$\frac{\Delta V_{BE}}{R_2} + \frac{V_{BE}}{R_1} = 1 \mu$$

Subs R_1 then $R_2 = 190 \text{ k}\Omega$
 $R_1 = 1.2 \text{ M}\Omega$
 $R_1 = R_0 = 1.2 \text{ M}\Omega$

For R_3 I pass $1 \mu\text{A}$ and need $v_{out} = 800 \text{ mV}$
So $R_{out} = 800 \text{ k}\Omega$

Figure 1: Hand calculations showing resistor ratios, PTAT/CTAT current balances, and baseline component sizing.

1.2 Resistor Sweeps and Core Sizing

Following the initial hand calculations, parametric sweeps were carried out on the PTAT resistor to cancel the first-order temperature coefficient around room temperature. The optimal zero-temperature coefficient point was obtained at:

- **PTAT Resistor** ($R_{PTAT} = R_2$): $175 \text{ k}\Omega$
- **Output Resistor** ($R_{out} = R_3$): $780 \text{ k}\Omega$ (establishing $V_{OUT} \approx 800 \text{ mV}$ at $I_{branch} \approx 1 \mu\text{A}$)

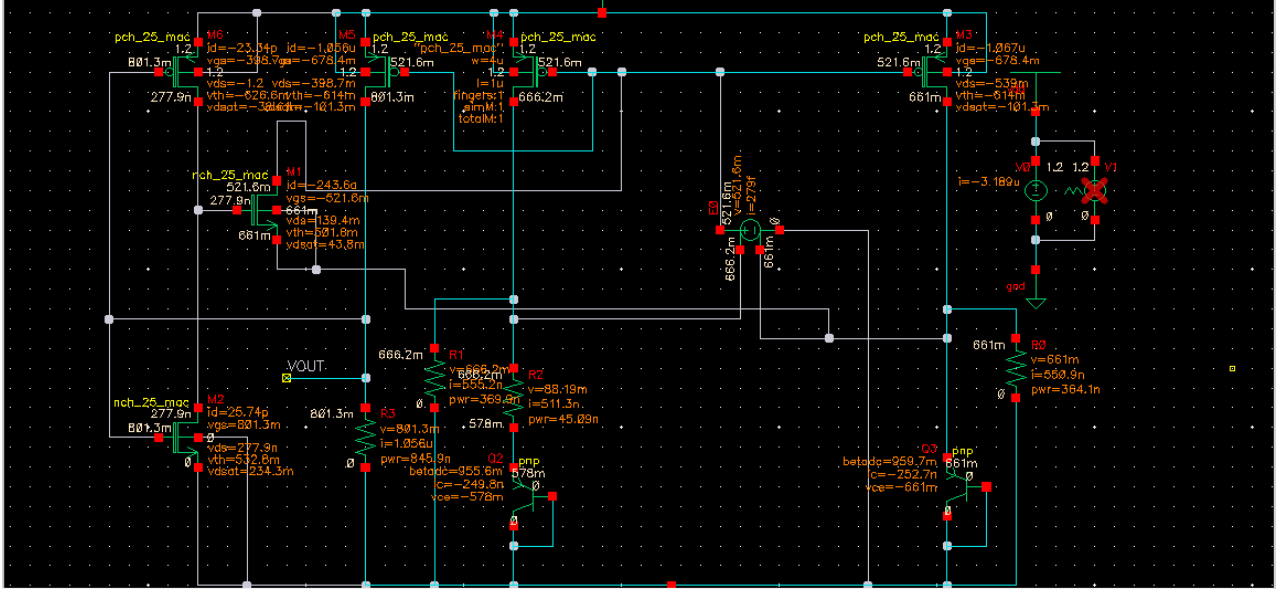


Figure 2: Core BGR circuit with startup circuit and behavioral VCVS amplifier showing DC operating points.

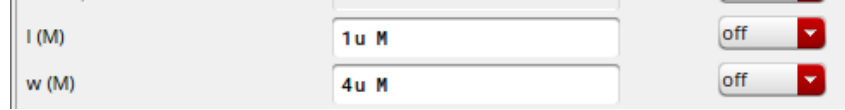


Figure 3: PMOS current mirror device sizing ($W = 4 \mu\text{m}$, $L = 1 \mu\text{m}$).

1.3 Operating Point (DCOP) Summary

Table 1: Part 1 Transistor Operating Point Summary

Device	Role	I_D	V_{GS}	V_{DS}	V_{TH}	V_{DSAT}
M_6	Startup PMOS	-23.34 pA	-398.7 mV	-1.200 V	-626.6 mV	-32.6 mV
M_2	Startup NMOS	25.74 pA	801.3 mV	277.9 mV	532.8 mV	234.3 mV
M_1	Startup NMOS	-243.6 aA	-521.6 mV	139.4 mV	501.8 mV	43.8 mV
M_5	Core PMOS	$-1.056 \mu\text{A}$	-678.4 mV	-398.7 mV	-614.0 mV	-101.3 mV
M_4	Core PMOS	$-1.056 \mu\text{A}$	-678.4 mV	-533.8 mV	-614.0 mV	-101.3 mV
M_3	Core PMOS	$-1.067 \mu\text{A}$	-678.4 mV	-539.0 mV	-614.0 mV	-101.3 mV

Table 2: Part 1 Passive Components and BJT Operating Points

Component	Voltage Drop	Branch Current	Power / β_{DC}
R_3 (R_{out})	801.3 mV	$1.056 \mu\text{A}$	845.9 nW
R_1	666.2 mV	555.2 nA	369.9 nW
R_2 (R_{PTAT})	88.19 mV	511.3 nA	45.09 nW
R_0	661.0 mV	550.9 nA	364.1 nW
Q_2	$V_{CE} = -578.0 \text{ mV}$	$I_C = -249.8 \text{ nA}$	$\beta_{DC} = 955.6 \text{ m}$
Q_3	$V_{CE} = -661.0 \text{ mV}$	$I_C = -252.7 \text{ nA}$	$\beta_{DC} = 959.7 \text{ m}$
Overall Circuit Performance: $V_{DD} = 1.20 \text{ V}$, $I_{total} = 3.189 \mu\text{A}$, $V_{OUT} = 801.3 \text{ mV}$			

1.4 Temperature Performance

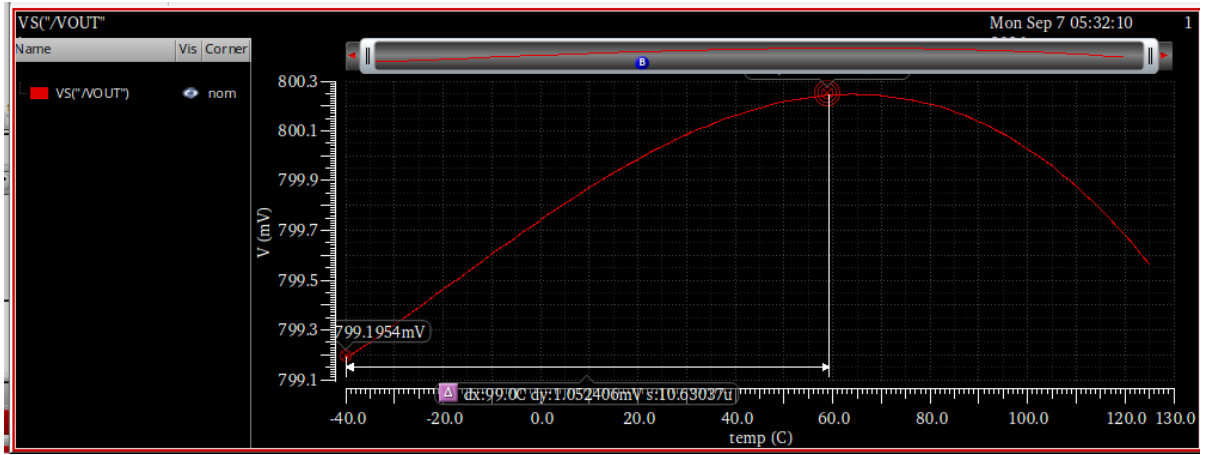


Figure 4: Nominal Corner (TT): V_{OUT} vs. Temperature (-40°C to $+130^{\circ}\text{C}$), demonstrating low temperature drift ($\Delta V_{OUT} \approx 1.052\text{ mV}$).

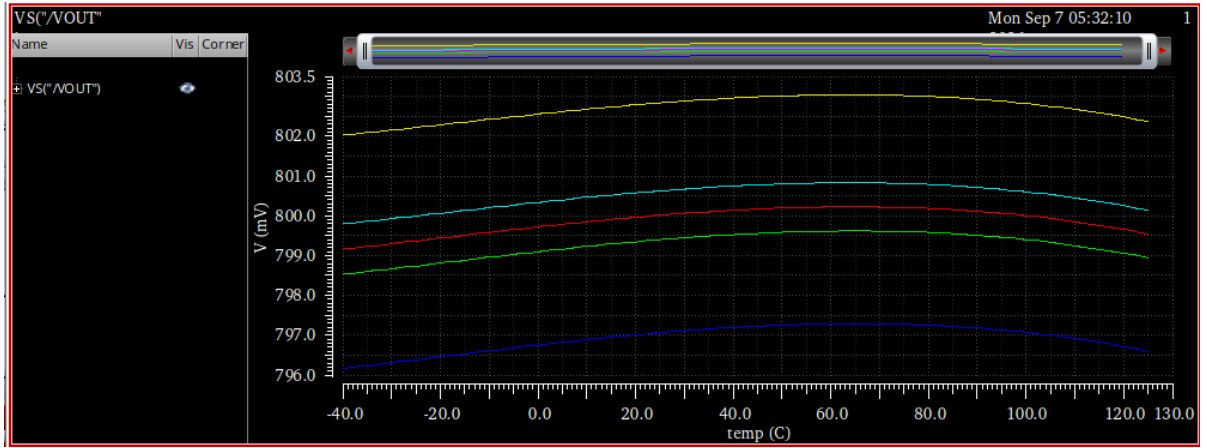


Figure 5: Process Corners (TT , SS , FF , SF , FS): V_{OUT} vs. Temperature with ideal VCVS.

2 Part 2: Integration of Self-Biased Amplifier

2.1 Amplifier Implementation and Resistor Retuning

Upon replacing the behavioral VCVS with the transistor-level self-biased OTA, the passive values were adjusted to compensate for finite amplifier gain, systemic offset, and input impedance:

- **PTAT Resistor** ($R_{PTAT} = R_2$): Retuned to 165 k Ω
- **Output Resistor** ($R_{out} = R_3$): Retuned to 770 k Ω

These tuned values restore the target output voltage to 801.3 mV while preserving optimal temperature curvature compensation.

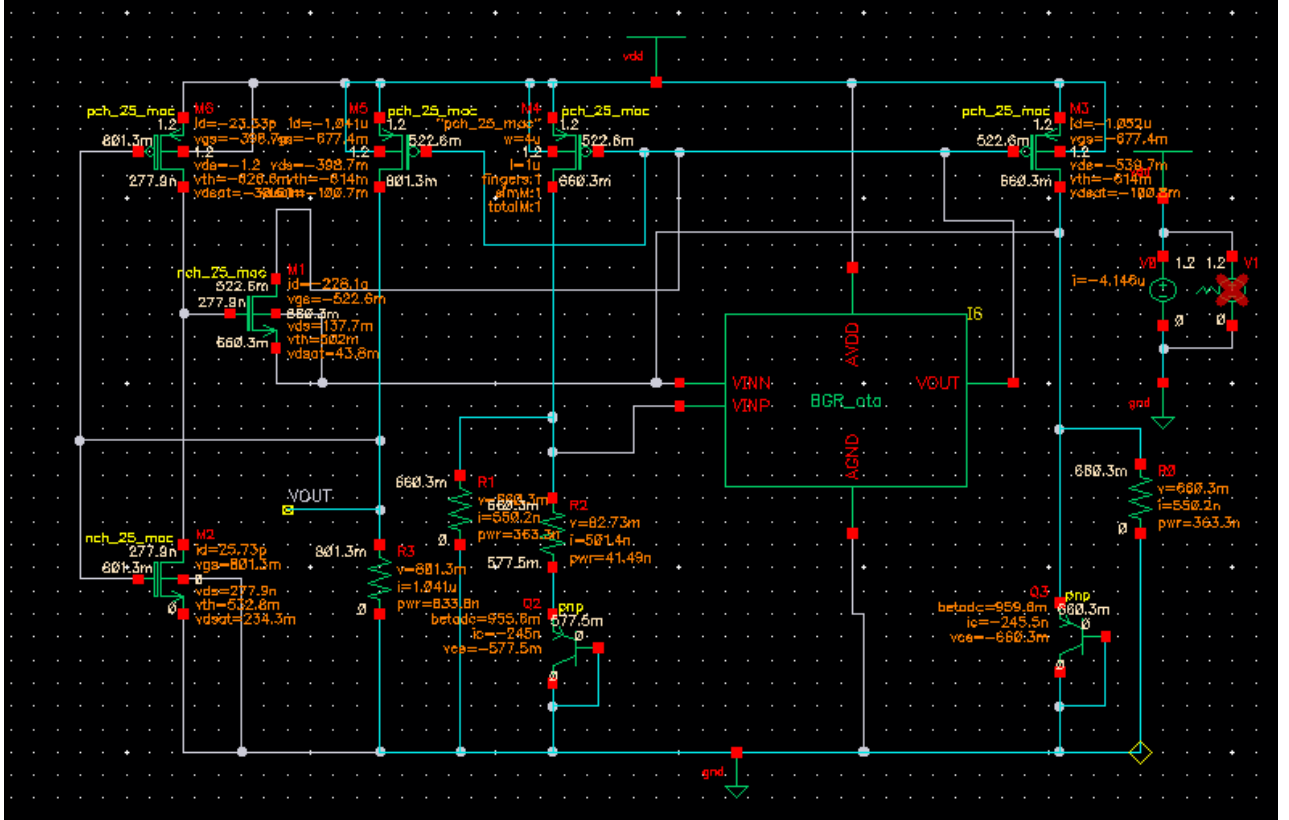


Figure 6: Top-level core integrated with self-biased OTA (BGR_ota) and startup circuit.

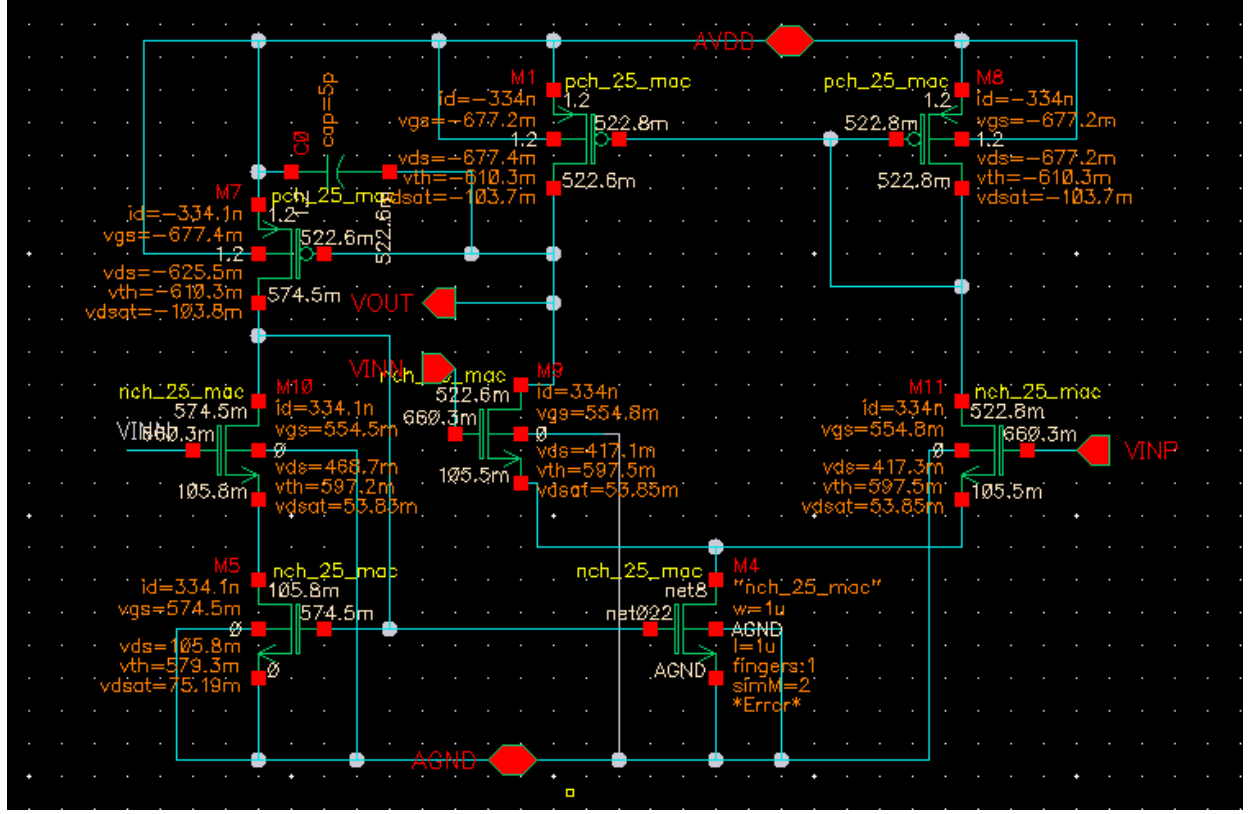


Figure 7: Internal schematic of the self-biased OTA (BGR_ota) with compensation capacitor ($C_0 = 5$ pF).

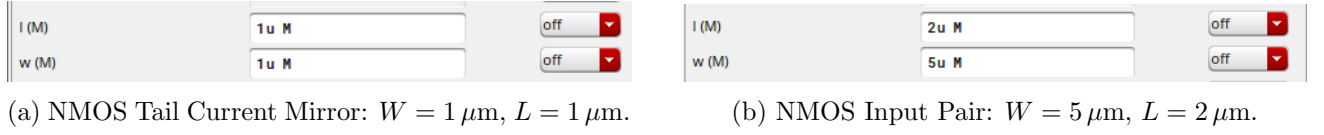


Figure 8: Transistor sizing for the self-biased operational transconductance amplifier.

2.2 Operating Point (DCOP) Summary with Self-Biased OTA

Table 3: Part 2 Complete Circuit Operating Point Data

Device	Block / Role	I_D	V_{GS}	V_{DS}	V_{TH}	V_{DSAT}
M_6	Startup PMOS	-23.33 pA	-398.7 mV	-1.200 V	-626.6 mV	-32.6 mV
M_2	Startup NMOS	25.73 pA	801.3 mV	277.9 mV	532.8 mV	234.3 mV
M_1	Startup NMOS	-228.1 aA	-522.6 mV	137.7 mV	502.0 mV	43.8 mV
M_5	Core PMOS	-1.041 μA	-677.4 mV	-398.7 mV	-614.0 mV	-100.7 mV
M_4	Core PMOS	-1.041 μA	-677.4 mV	-533.7 mV	-614.0 mV	-100.7 mV
M_3	Core PMOS	-1.052 μA	-677.4 mV	-539.7 mV	-614.0 mV	-100.8 mV
M_7	OTA PMOS Load	-334.1 nA	-677.4 mV	-625.5 mV	-610.3 mV	-103.8 mV
M_1	OTA PMOS Load	-334.0 nA	-677.2 mV	-677.4 mV	-610.3 mV	-103.7 mV
M_8	OTA PMOS Load	-334.0 nA	-677.2 mV	-677.2 mV	-610.3 mV	-103.7 mV
M_{10}	OTA NMOS Pair	334.1 nA	554.5 mV	468.7 mV	597.2 mV	53.83 mV
M_9	OTA NMOS Pair	334.0 nA	554.8 mV	417.1 mV	597.5 mV	53.85 mV
M_{11}	OTA NMOS Pair	334.0 nA	554.8 mV	417.3 mV	597.5 mV	53.85 mV
M_5	OTA Tail Diode	334.1 nA	574.5 mV	105.8 mV	579.3 mV	75.19 mV
Performance: $V_{DD} = 1.20$ V, $I_{total} = 4.146 \mu\text{A}$ ($< 10 \mu\text{A}$ spec), $V_{OUT} = 801.3$ mV.						

2.3 Stability Analysis (STB) and Phase Margin

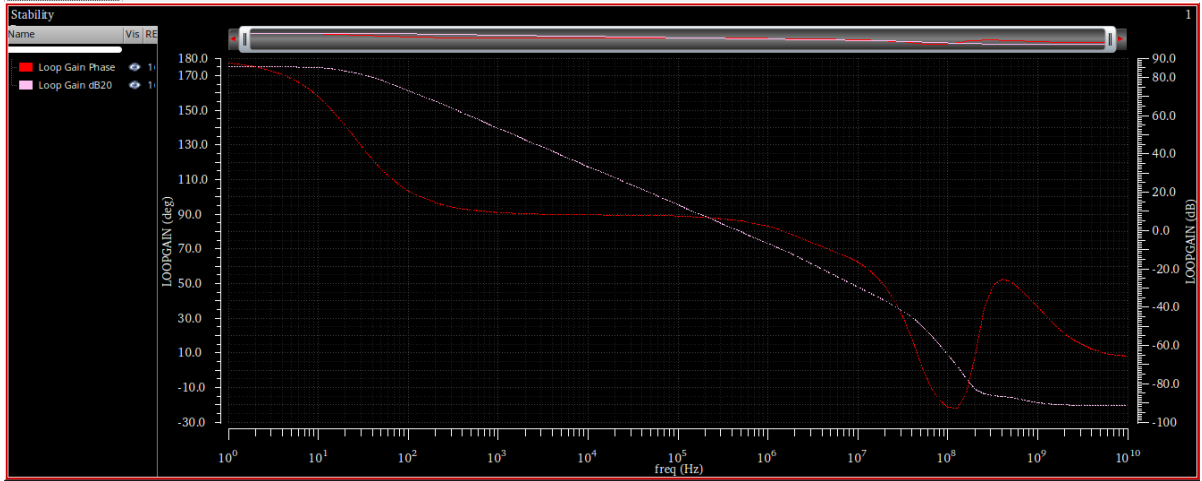


Figure 9: Loop gain magnitude (dB) and phase ($^{\circ}$) frequency response from the STB simulation.

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BGR:BGRideal:1 (phaseMargin(getData("loopGai... 86.71
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Figure 10: Cadence ADE simulation output expression verifying a Phase Margin of 86.71° .

Table 4: Stability Performance Summary

Parameter	Target Specification	Simulated Value	Status
Phase Margin (PM)	$> 60.0^{\circ}$	86.71°	Pass

2.4 Transient Startup Simulation

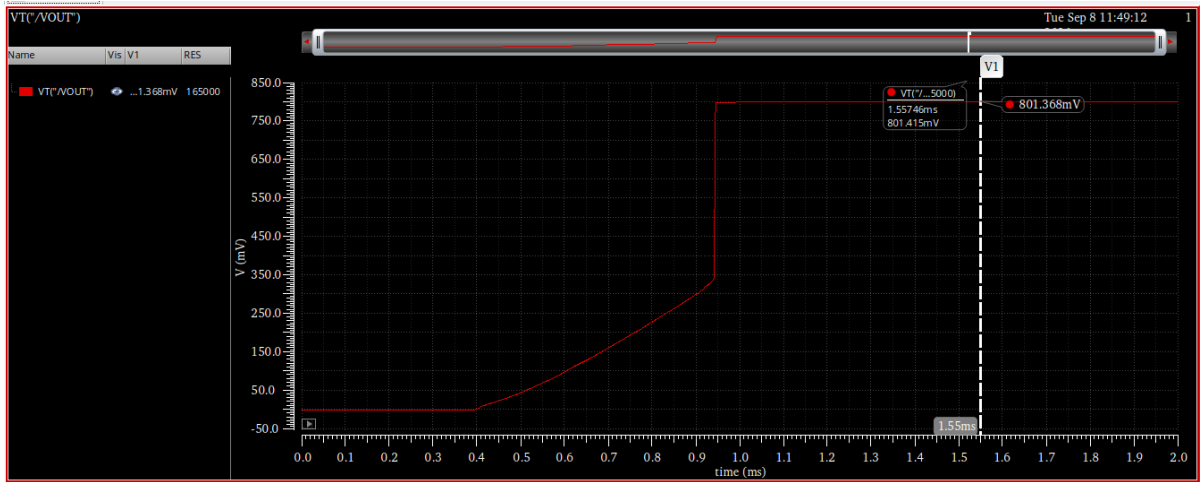


Figure 11: Transient startup of V_{OUT} under a 1 ms supply ramp, settling cleanly to 801.368 mV at $t = 1.55$ ms.

2.5 Corner Temperature Variations

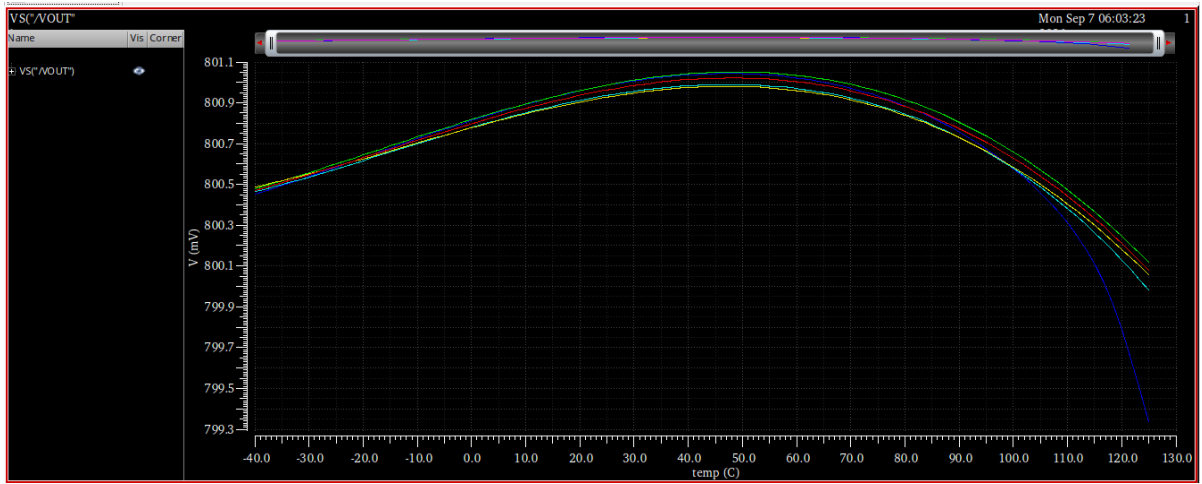


Figure 12: Temperature sweep (-40°C to $+125^{\circ}\text{C}$) across process corners (TT , SS , FF , SF , FS) with the self-biased OTA.

3 Part 3 (Bonus): Implementation with Actual PDK Passives

3.1 Selected Passives and Device Sizing

Foundry PDK components replace all ideal elements:

- **Resistors:** Selected `rppolywo_mx` (unsalicyded P-poly) with fixed width $W = 400$ nm. Lengths are adjusted to match target resistances (≈ 1.278 M Ω).
- **Capacitors:** PDK passive capacitor model integrated for frequency compensation ($C_0 = 5$ pF).

3.2 Complete Schematic with Annotated Operating Points

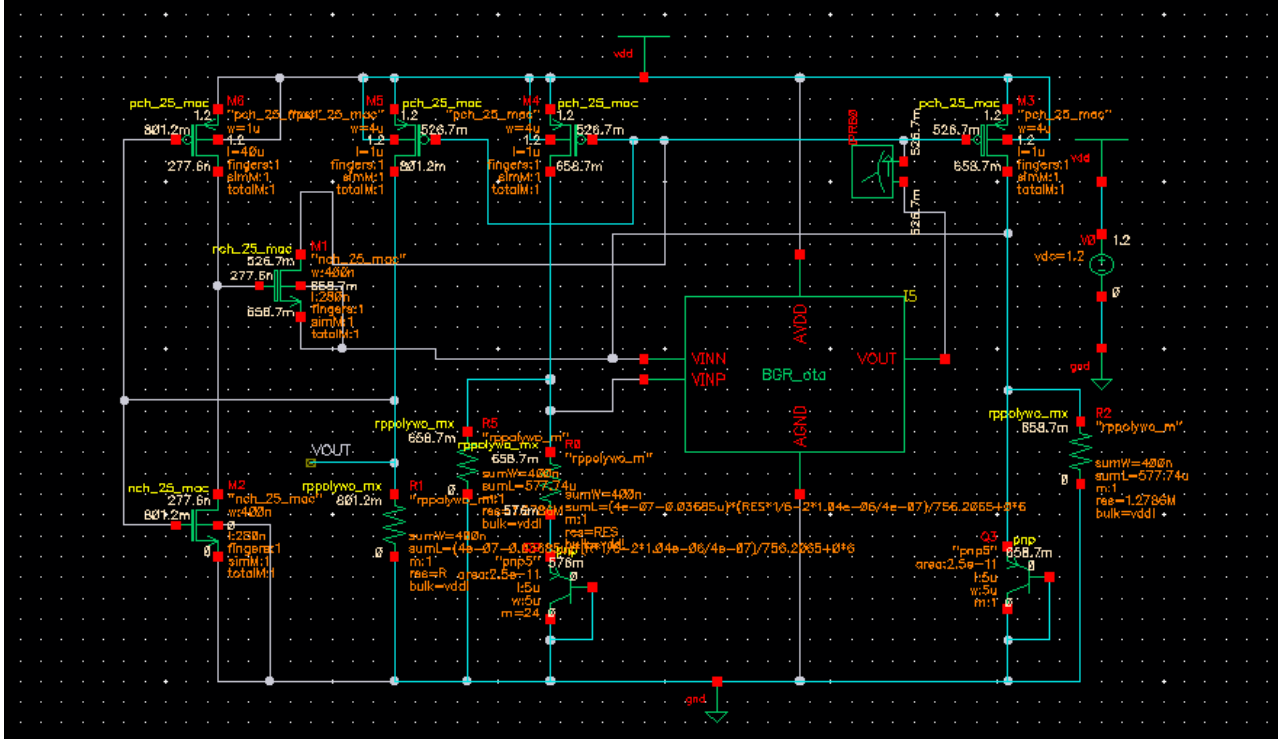


Figure 13: Complete schematic using PDK `rppolywo_mx` resistors and `pnp5` BJTs with annotated node voltages and DC operating points.

3.3 Operating Point (DCOP) Summary from Cadence Simulation Dump

Table 5: Part 3 Core Transistor Operating Point Data

Device	Role	I_D	V_{GS}	V_{DS}	V_{TH}	V_{DSAT}
M_5	Core PMOS Mirror	-977.2 nA	-673.3 mV	-398.8 mV	-614.0 mV	-98.38 mV
M_4	Core PMOS Mirror	-987.7 nA	-673.3 mV	-541.3 mV	-614.0 mV	-98.38 mV
M_3	Core PMOS Mirror	-987.7 nA	-673.3 mV	-541.3 mV	-614.0 mV	-98.38 mV

Table 6: Part 3 Amplifier (BGR_ota) Internal Transistor Operating Point Data

Device	Role	I_D	V_{GS}	V_{DS}	V_{TH}	V_{DSAT}
I_5/M_7	PMOS Active Load	-314.1 nA	-673.3 mV	-629.0 mV	-610.3 mV	-101.4 mV
I_5/M_1	PMOS Active Load	-314.0 nA	-673.1 mV	-673.3 mV	-610.3 mV	-101.3 mV
I_5/M_8	PMOS Active Load	-314.0 nA	-673.1 mV	-673.1 mV	-610.3 mV	-101.3 mV
I_5/M_{11}	NMOS Input Pair	314.0 nA	552.3 mV	420.4 mV	597.7 mV	53.10 mV
I_5/M_9	NMOS Input Pair	314.0 nA	552.3 mV	420.3 mV	597.7 mV	53.10 mV
I_5/M_{10}	NMOS Input Pair	314.1 nA	552.0 mV	464.2 mV	597.5 mV	53.08 mV
I_5/M_4	Tail Current Sink	628.0 nA	571.0 mV	106.4 mV	579.2 mV	73.71 mV
I_5/M_5	Tail Diode-Ref	314.1 nA	571.0 mV	106.7 mV	579.2 mV	73.71 mV

3.4 Temperature Performance across Process Corners (Part 3)

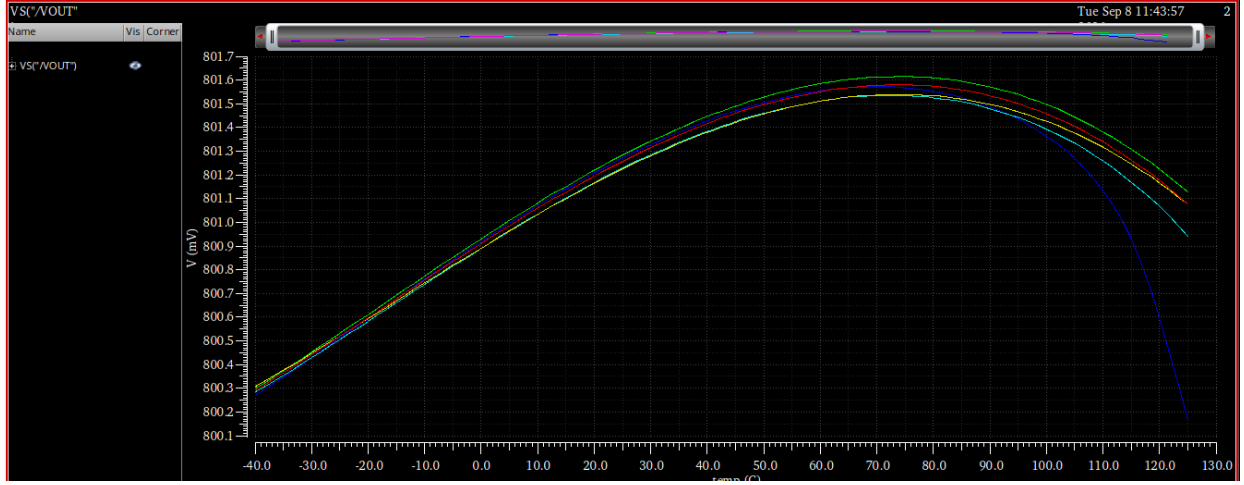


Figure 14: V_{OUT} vs. Temperature (-40°C to $+130^{\circ}\text{C}$) across corners (TT , SS , FF , SF , FS) using actual PDK rppolywo_mx passives.